

# PROJET : Automatisation d'une Infrastructure Web Multi-OS



**MAYITOUKOU SIGMUND**

INGÉNIEUR DEVOPS

# ÉNONCÉ DU PROJET

## Contexte réel

Vous êtes DevOps Engineer dans une entreprise qui souhaite standardiser le provisioning de ses serveurs.

L'entreprise possède :

- 1 serveur Ubuntu (héberge application web)
- 1 serveur Fedora (serveur API interne)
- 1 serveur Windows (serveur IIS pour application legacy)

Votre mission est d'automatiser entièrement la configuration des machines avec Ansible.

## Objectifs du projet

Automatiser :

### 1 Sur Ubuntu et Fedora

- Mise à jour du système
- Installation :
  - Apache (apache2 ou httpd)
  - PHP
  - MariaDB
  - Git
  - Curl
- Création d'un utilisateur **devops**
- Déploiement d'une application PHP
- Sécurisation MariaDB
- Création base de données **company\_db**
- Création utilisateur MySQL **company\_user**
- Activation et démarrage des services

- Firewall ouvert sur port 80

## 2 Sur Windows Server

- Installation IIS
- Démarrage service W3SVC
- Création dossier C:\CompanyApp
- Déploiement fichier index.html
- Activation Firewall port 80

## 3 Contraintes

- Utiliser des rôles
- Séparer Linux et Windows
- Utiliser handlers
- Utiliser variables dynamiques selon OS
- Idempotence obligatoire
- Mots de passe via variables (bonus : Vault)

## Structure attendue

```
automation-project/
|
|— inventory/
|   |— hosts.ini
|
|— site.yml
|
|— roles/
|   |— linux_common/
|   |— database/
|   |— webserver/
|   |— windows_iis/
```

## CORRIGÉ COMPLET

## inventory/hosts.ini

```
[ubuntu]  
192.168.1.10
```

```
[fedora]  
192.168.1.11
```

```
[linux:children]  
ubuntu  
fedora
```

```
[windows]  
192.168.1.20
```

```
[linux:vars]  
ansible_user=devops  
ansible_become=true
```

```
[windows:vars]  
ansible_user=Administrator  
ansible_password=Password123!  
ansible_connection=winrm  
ansible_winrm_transport=ntlm
```

## site.yml

```
- name: Configuration serveurs Linux  
  hosts: linux  
  become: yes  
  roles:  
    - linux_common  
    - webserver  
    - database  
  
- name: Configuration serveur Windows  
  hosts: windows
```

roles:

- windows\_iis

## **ROLE 1 : linux\_common**

### **tasks/main.yml**

- name: Mise à jour Debian

apt:

update\_cache: yes

when: ansible\_os\_family == "Debian"

- name: Mise à jour Fedora

dnf:

update\_cache: yes

when: ansible\_os\_family == "RedHat"

- name: Créer utilisateur devops

user:

name: devops

groups: sudo

shell: /bin/bash

state: present

## **ROLE 2 : webserver**

### **vars/main.yml**

apache\_service: "{{ 'apache2' if ansible\_os\_family == 'Debian' else 'httpd' }}"

debian\_packages:

- apache2

- php
- libapache2-mod-php
- git
- curl

redhat\_packages:

- httpd
- php
- git
- curl

## **tasks/main.yml**

- name: Installer paquets Debian

apt:

name: "{{ debian\_packages }}"

state: present

when: ansible\_os\_family == "Debian"

- name: Installer paquets Fedora

dnf:

name: "{{ redhat\_packages }}"

state: present

when: ansible\_os\_family == "RedHat"

- name: Démarrer Apache

service:

name: "{{ apache\_service }}"

state: started

enabled: yes

- name: Ouvrir firewall port 80

firewalld:

port: 80/tcp

permanent: yes

state: enabled

when: ansible\_os\_family == "RedHat"

- name: Déployer app PHP  
copy:  
 src: index.php  
 dest: /var/www/html/index.php  
notify: restart apache

### **handlers/main.yml**

- name: restart apache  
service:  
 name: "{{ apache\_service }}"  
 state: restarted

## **ROLE 3 : database**

### **vars/main.yml**

db\_root\_password: RootPass123!  
db\_name: company\_db  
db\_user: company\_user  
db\_password: Company123!

### **tasks/main.yml**

- name: Installer MariaDB Debian  
apt:  
 name:  
 - mariadb-server  
 - python3-pymysql  
 state: present  
when: ansible\_os\_family == "Debian"

- name: Installer MariaDB Fedora  
dnf:

```

name:
  - mariadb-server
  - python3-PyMySQL
state: present
when: ansible_os_family == "RedHat"

- name: Démarrer MariaDB
service:
  name: mariadb
  state: started
  enabled: yes

- name: Créer base
mysql_db:
  name: "{{ db_name }}"
  state: present
  login_user: root
  login_password: "{{ db_root_password }}"

- name: Créer utilisateur DB
mysql_user:
  name: "{{ db_user }}"
  password: "{{ db_password }}"
  priv: "{{ db_name }}.*:ALL"
  state: present
  login_user: root
  login_password: "{{ db_root_password }}"

```

## **ROLE 4 : windows\_iis**

### **tasks/main.yml**

```

- name: Installer IIS
win_feature:
  name: Web-Server
  state: present

```

- name: Démarrer IIS
  - win\_service:
    - name: W3SVC
    - state: started
  
- name: Créer dossier application
  - win\_file:
    - path: C:\CompanyApp
    - state: directory
  
- name: Déployer page HTML
  - win\_copy:
    - content: "<h1>Company Windows Server</h1>"
    - dest: C:\CompanyApp\index.html
  
- name: Ouvrir firewall port 80
  - win\_firewall\_rule:
    - name: HTTP
    - localport: 80
    - action: allow
    - direction: in
    - protocol: TCP